



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

SpaceX Falcon 9 First
Stage Landing
Prediction
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Executive Summary

Data: 90 historical Falcon 9 launches (2010–2020); 60 successful first-stage landings (66.7%).

EDA: KSC LC 39A reached 17/22 (77.3%) successful landings; CCAFS SLC 40 had 33/55 (60.0%). VLEO: 12/14 (85.7%); GTO: 14/27 (51.9%).

Prediction: decision tree, logistic regression and SVM each scored 14/18 (77.8%) on the held-out test set. The tree led five-fold validation at 87.6%.

Interpretation: later launches show higher landing rates in this sample. Small groups and 18 test cases limit certainty; associations are not causal.

Introduction

Problem: predicting successful recovery of the Falcon 9 first stage can inform launch-cost estimates for a competing provider.

Question: how do launch site, orbit, payload, flight number and booster features relate to landing success?

Scope: IBM course snapshots of historical launches; Class 1 means successful landing and Class 0 means no successful landing. This report does not describe current operations.

Section 1

Methodology

Methodology

Collect IBM course launch data and inspect missing values.

Summarize landing by site, orbit, payload and time.

Use SQLite for the supplied SQL exercise dataset.

Create site views and compare four classifiers

Data Collection

Analysis uses four IBM course datasets: 90 launch records, 101 SQL exercise records, 56 dashboard records and 90 encoded feature rows. These samples differ; comparisons always state the source.

Data Collection – SpaceX API

Course snapshot of SpaceX API
launch data → Falcon 9 filter →
enrich rocket, payload, launchpad
and core IDs → 90 launch records.

The course snapshot makes
the analysis reproducible.

API snapshot



Falcon 9 filter



Resolve IDs



Landing data

Data Collection - Scraping

The course scraping task extracts Falcon 9 launch rows from the Wikipedia launch history and standardizes dates, boosters and payloads. Here, the published IBM SQL CSV is used for reproducible query results; this report does not claim a fresh scrape.

<https://github.com/delt-kz/ibm-spacex-capstone-2026>

Launch tables



Parse rows



Normalize fields



SQL dataset

Data Wrangling

- The 90-row launch dataset has 26 missing LandingPad entries. Keep them as missing: no pad may mean no planned pad landing. Use the provided Class label, inspect categories, and one-hot encoded features for models. No outcome rows are removed.

EDA with Data Visualization

Charts compare landing outcome by launch site, orbit, flight number, payload mass and year. Group sizes are shown because rates from one or two launches are unstable.

EDA with SQL

SQLite queries reproduce the ten course tasks: unique sites, CCA launch rows, NASA payload, F9 v1.1 average payload, first ground landing, drone-ship cases, mission outcomes, maximum payload, 2015 failures and outcome ranking.

Build an Interactive Map with Folium

Plot the three distinct launch coordinates as markers. Color launch records by Class, then inspect site proximity. The course dataset gives coordinates but no rail, highway or coastline geometry; those distances are not inferred.

Build a Dashboard with Plotly Dash

Use the 56-row dashboard dataset for site-level pie charts and payload-versus-outcome exploration. Its rates differ from the 90-row API cohort, so the dashboard sample is labelled separately.

<https://github.com/delt-kz/ibm-spacex-capstone-2026>

Predictive Analysis (Classification)

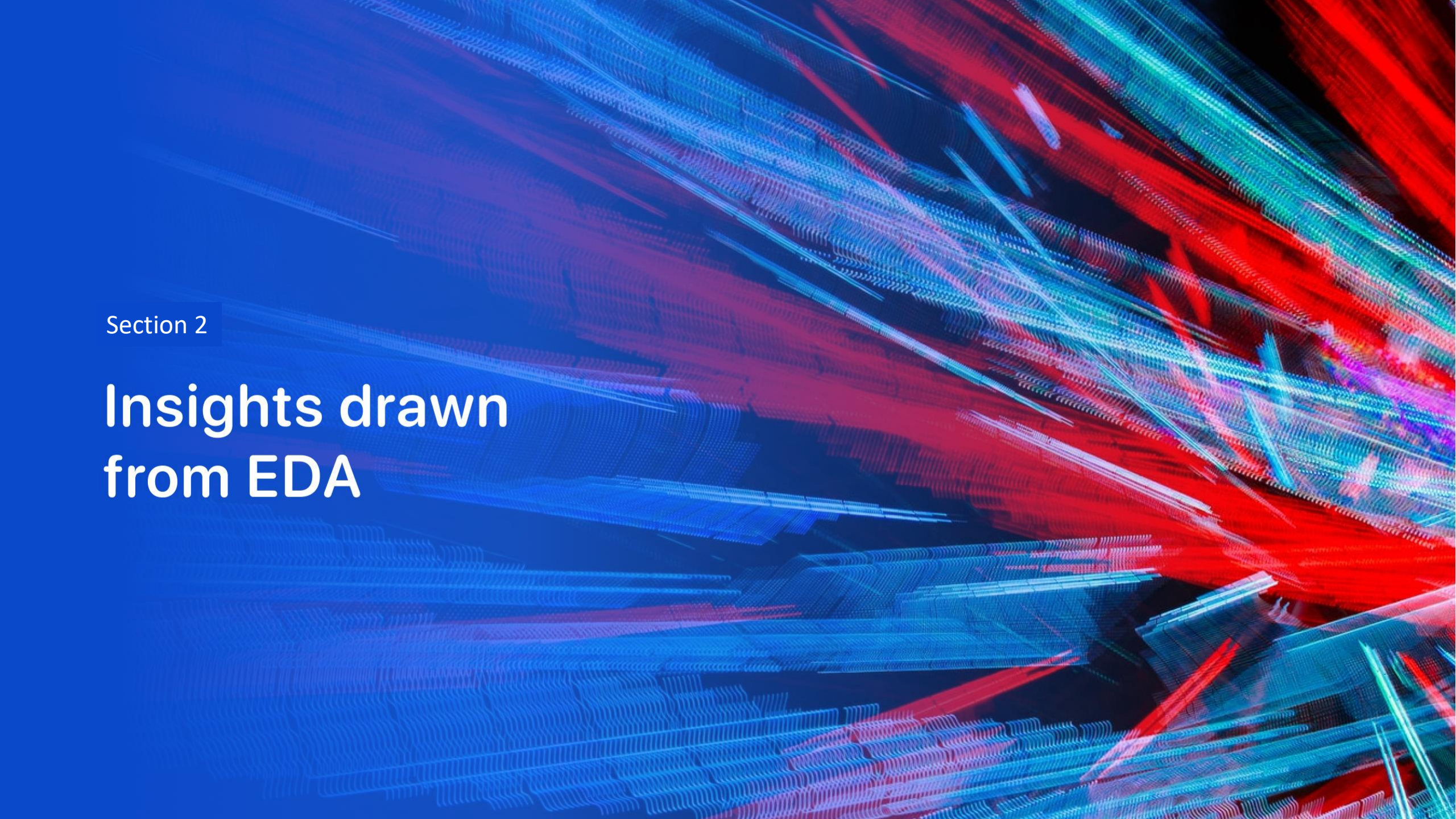
Features: IBM's 90-row one-hot encoded table; outcome: Class from the aligned launch dataset. FlightNumber is excluded from the feature matrix.

Split: stratified 72 training / 18 test launches (20% holdout; seed 2). StandardScaler is fit only on training rows.

Tuning: five-fold GridSearchCV compares logistic regression, RBF/linear SVM, depth-limited decision tree and K-nearest neighbors. The untouched test set is used once for final accuracy and confusion matrices.

Results

The 90-launch cohort contains 60 successful landings. Site success rates range from 60.0% to 77.3%. The best model reaches 77.8% held-out accuracy; the test set has only 18 launches.

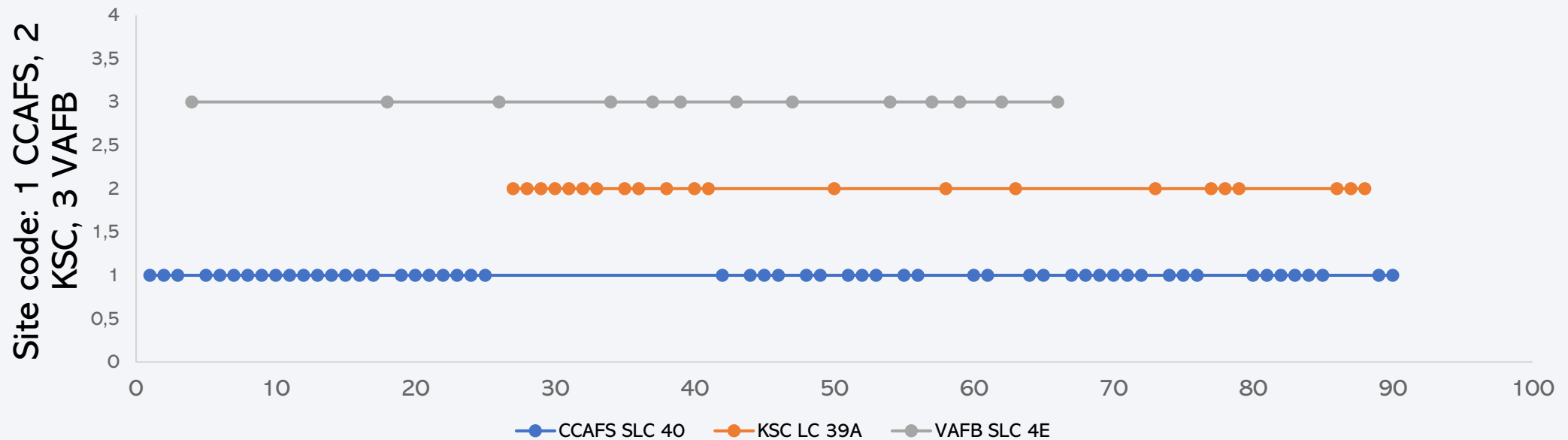
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

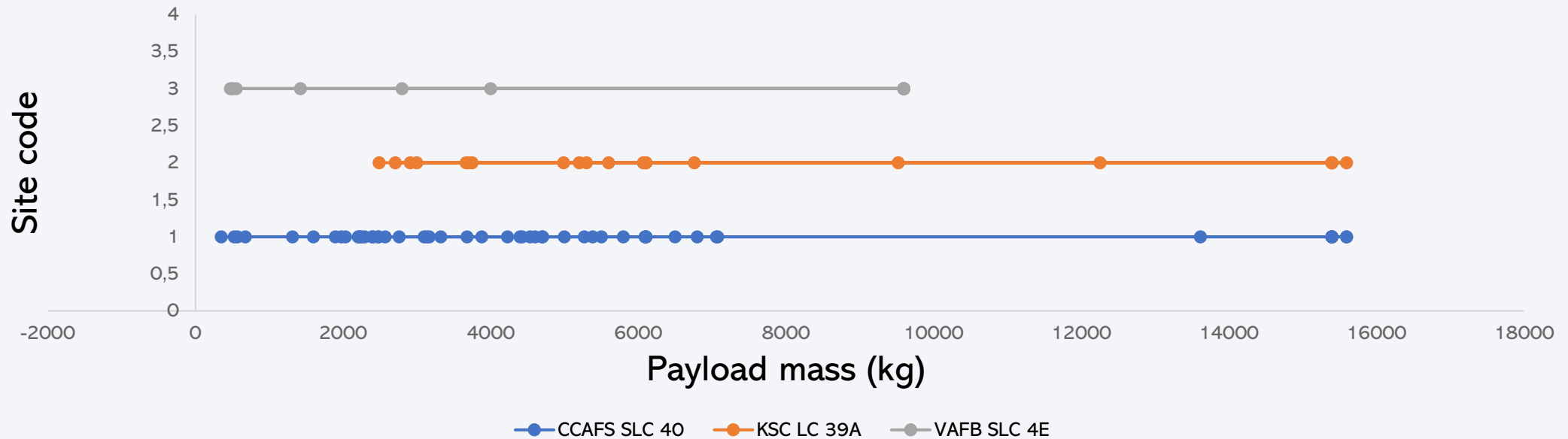
Flight Number vs. Launch Site

The 90 launches span three sites; CCAFS SLC 40 accounts for 55.



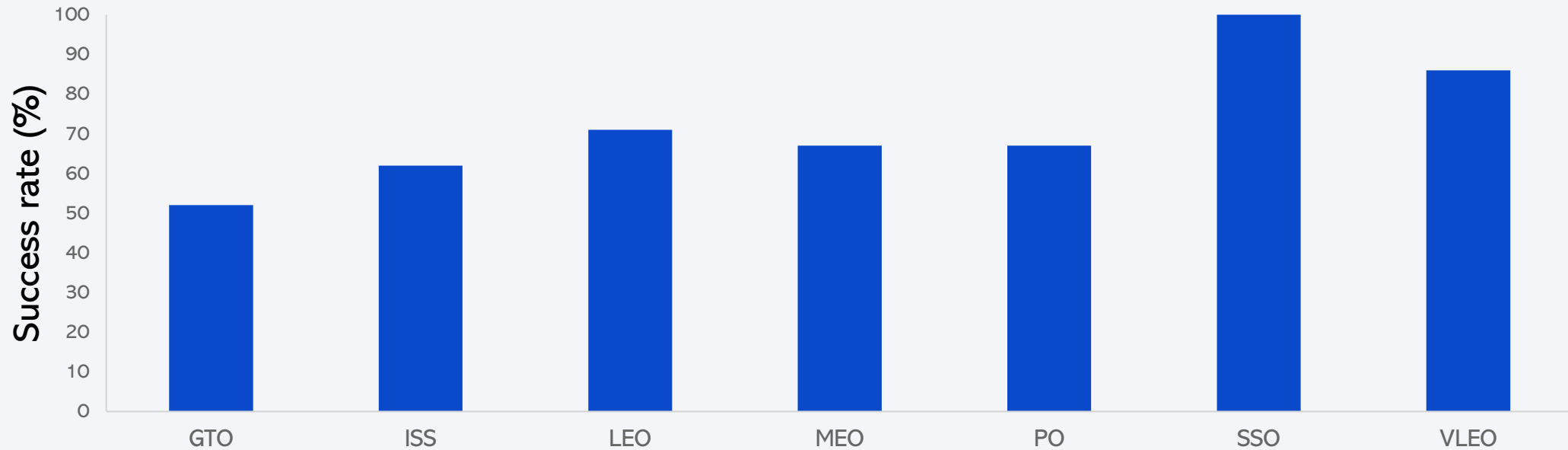
Payload vs. Launch Site

Payload mass ranges overlap across all three sites.



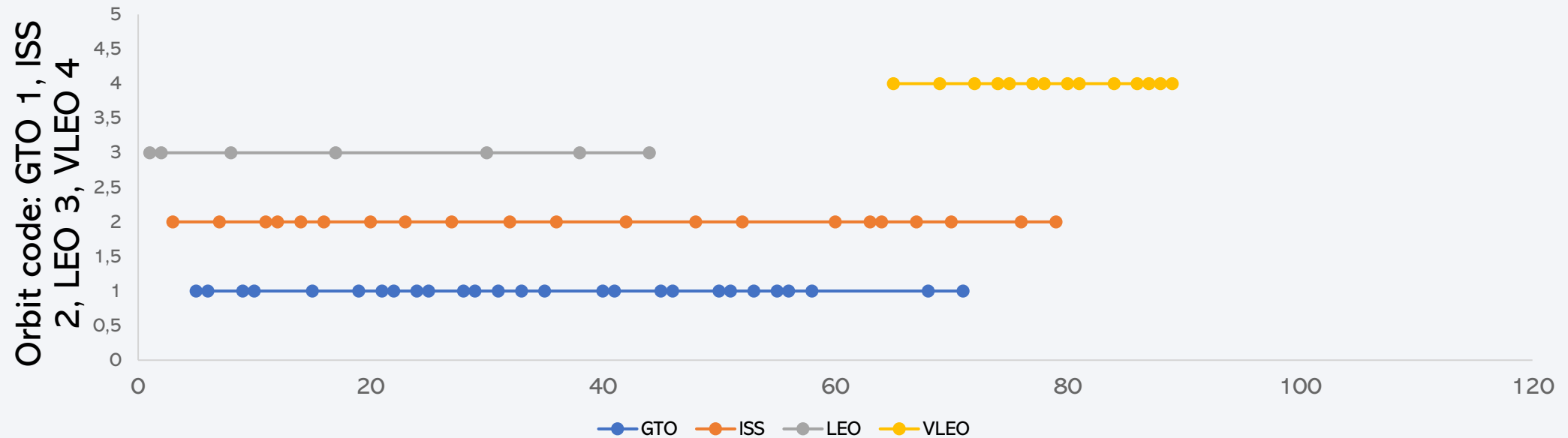
Success Rate vs. Orbit Type

VLEO has 12/14 successful landings (85.7%); GTO has 14/27 (51.9%).



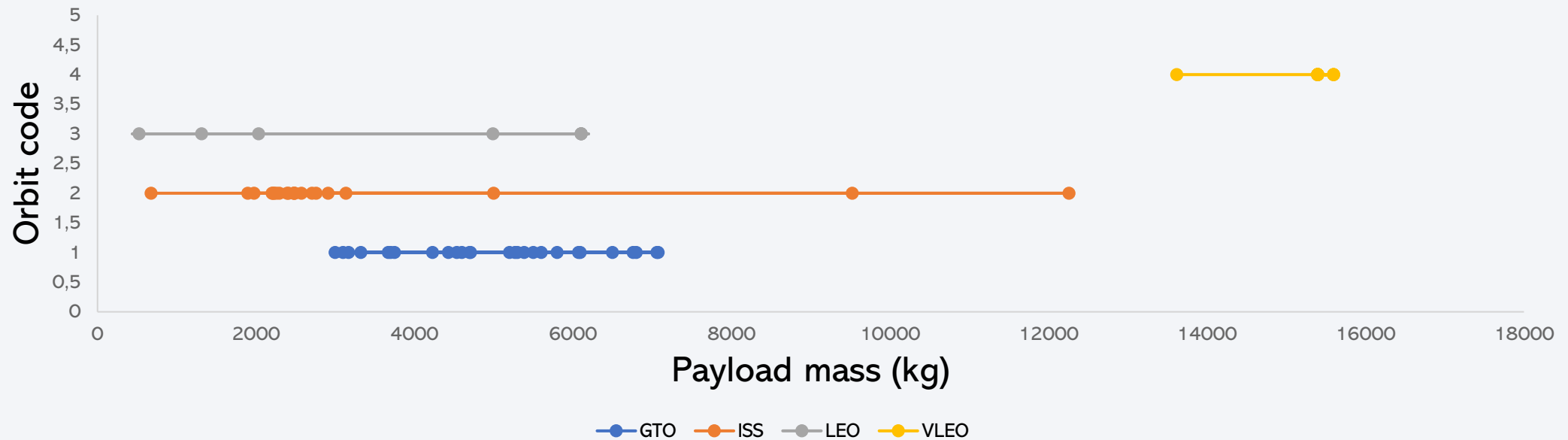
Flight Number vs. Orbit Type

Flight number approximates chronology; the orbit mix changes over time.



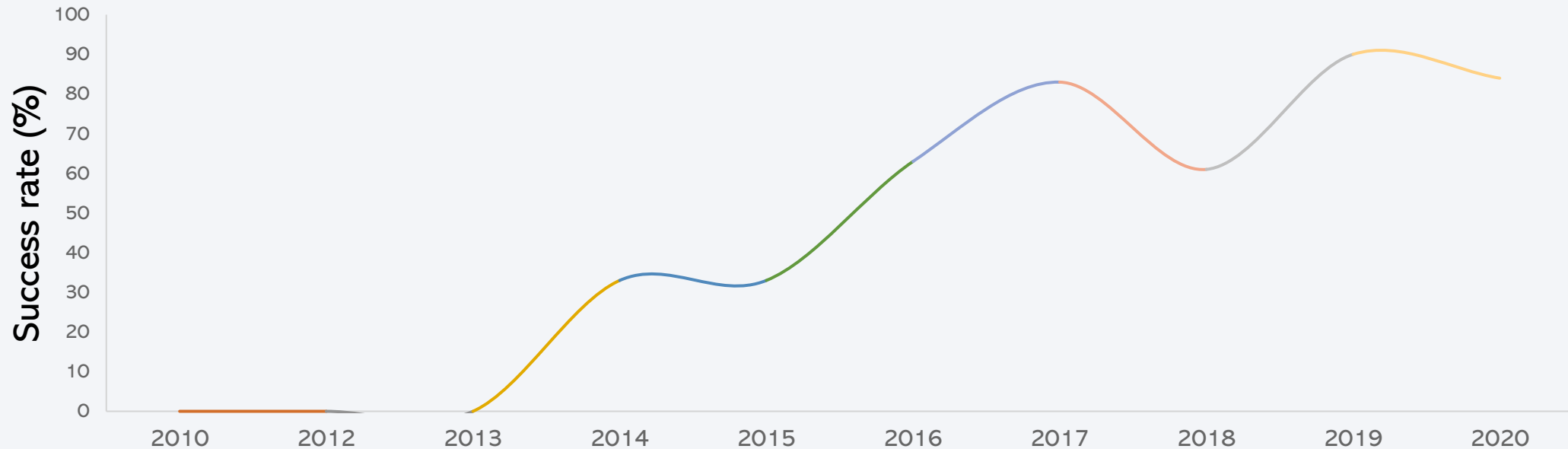
Payload vs. Orbit Type

Payload masses overlap across the four most common orbits.



Launch Success Yearly Trend

Yearly landing rates rise, but early years have small samples.



All Launch Site Names

Four distinct labels in the SQL sample:

CCAFS LC-40

CCAFS SLC-40

KSC LC-39A

VAFB SLC-4E

LC-40 and SLC-40 are separate labels in the source.

Launch Site Names Begin with 'CCA'

First five CCA rows:

2010-06-04 — CCAFS LC-40

2010-12-08 — CCAFS LC-40

2012-05-22 — CCAFS LC-40

2012-10-08 — CCAFS LC-40

2013-03-01 — CCAFS LC-40

Total Payload Mass

NASA-tagged customer rows carry 107 010 kg in total. Filter: Customer LIKE '%NASA%'.

Average Payload Mass by F9 v1.1

Average payload for F9 v1.1 variants: 2 534,7 kg. Filter: Booster_Version LIKE 'F9 v1.1%'.

First Successful Ground Landing Date

The first recorded successful ground-pad landing is 2015-12-22 in the SQL sample.

Successful Drone Ship Landing with Payload between 4000 and 6000

Drone-ship landings with 4,000–6,000 kg payload:

F9 FT B1021.2

F9 FT B1031.2

F9 FT B1022

F9 FT B1026

Total Number of Successful and Failure Mission Outcomes

Mission outcomes in 101 SQL rows:

Success: 98

Success (payload status unclear): 1

Success: 1

Failure (in flight): 1

The source has two differently spaced “Success” labels.

Boosters Carried Maximum Payload

Maximum payload is 15 600 kg. It occurs in 12 records across multiple F9 Block 5 boosters.

2015 Launch Records

Two 2015 drone-ship failures:

2015-01-10 — F9 v1.1 B1012 — CCAFS LC-40

2015-04-14 — F9 v1.1 B1015 — CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing outcomes through 2017-03-20:

No attempt: 10

Success (drone ship): 5

Failure (drone ship): 5

Success (ground pad): 3

Controlled (ocean): 3

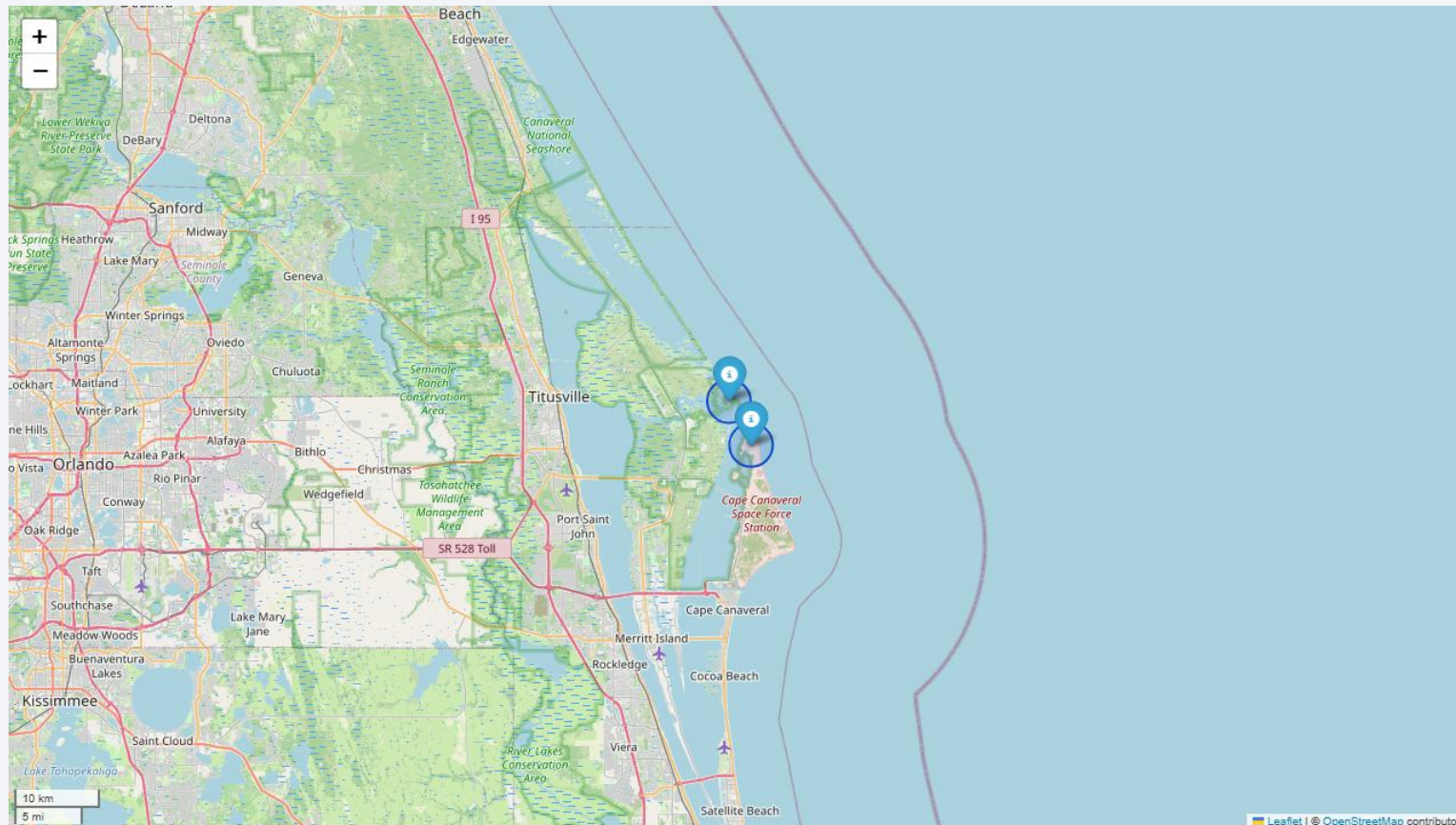
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left shows a clear blue sky.

Section 3

Launch Sites Proximities Analysis

Folium: Florida launch sites

OpenStreetMap basemap; circles mark the two Florida pads in the 90-launch cohort.



KSC LC 39A

17/22 successful

77.3%

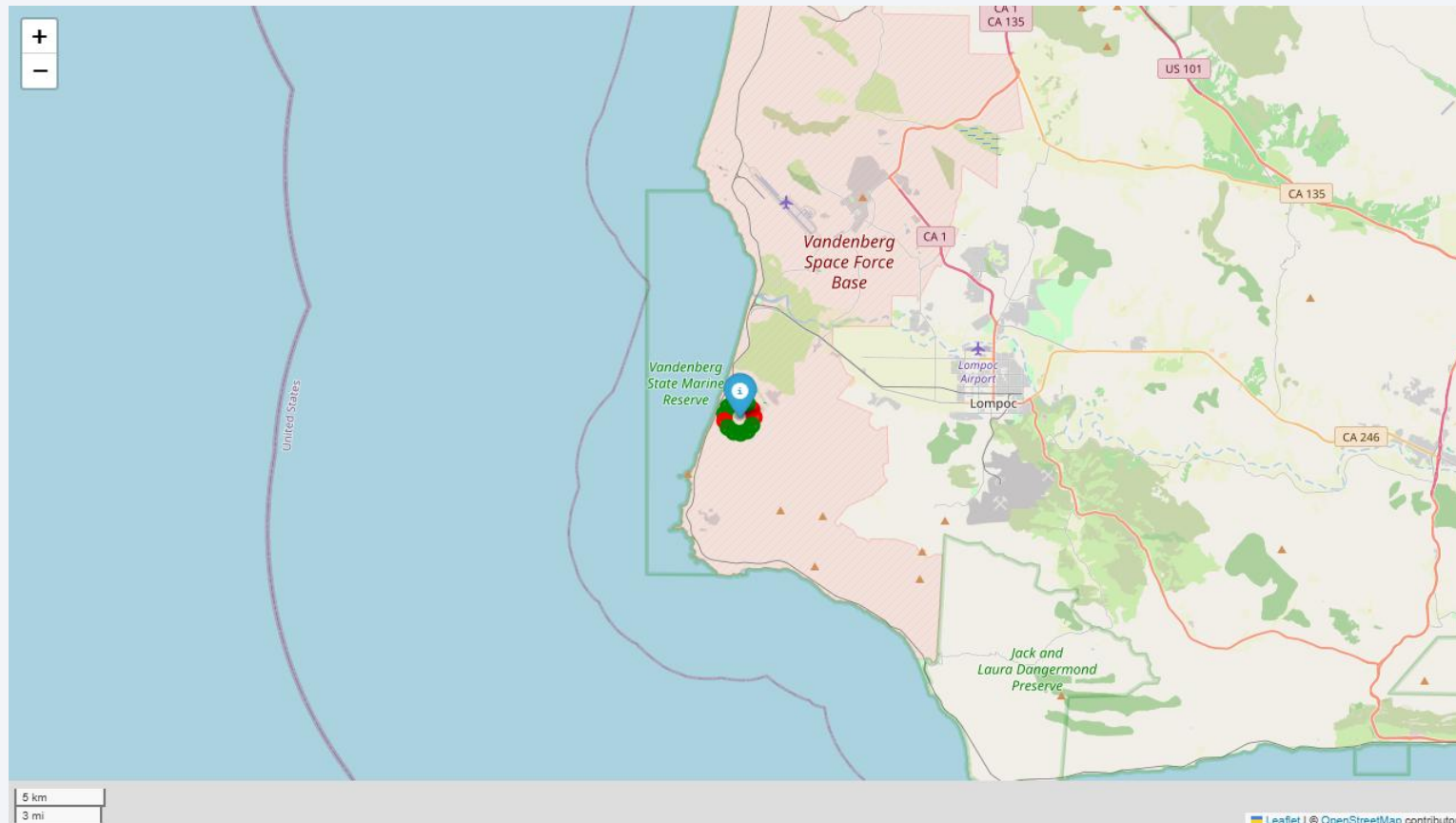
CCAFS SLC 40

33/55 successful

60.0%

Folium: VAFB landing outcomes

Green = successful landing; red = unsuccessful. Markers are slightly offset to reveal records at one pad.



VAFB SLC 4E

13 launches

10 successful

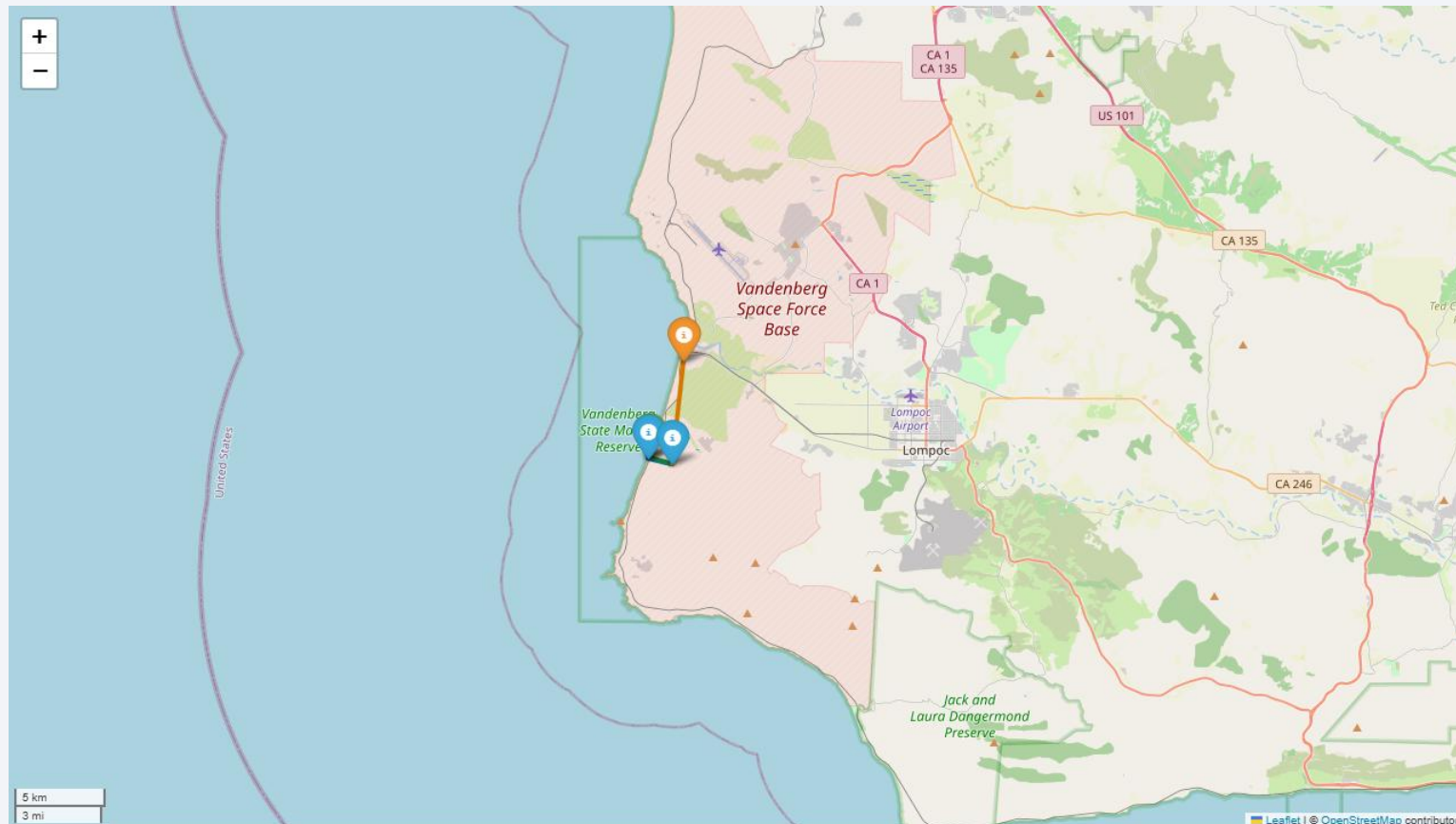
3 unsuccessful

True site coordinate:

34.632°N, 120.611°W

Folium: proximity near VAFB

Straight-line distances from VAFB SLC 4E to mapped features; these are not road travel distances.



Coastline: 1.35 km

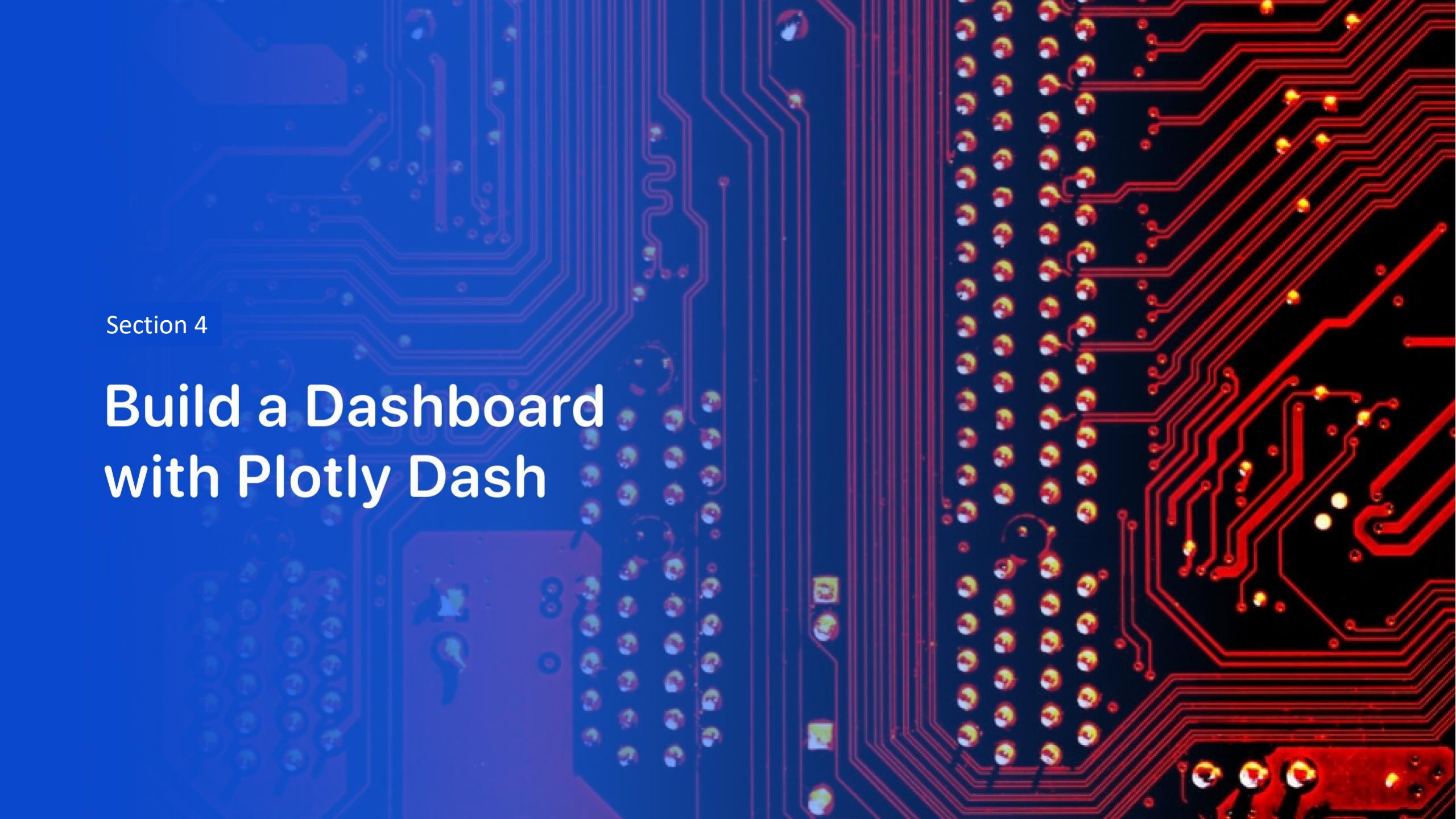
Main railway: 1.27 km

Ocean Ave road: 5.63 km

OpenStreetMap data

via Overpass API

snapshot: 28 Jul 2026

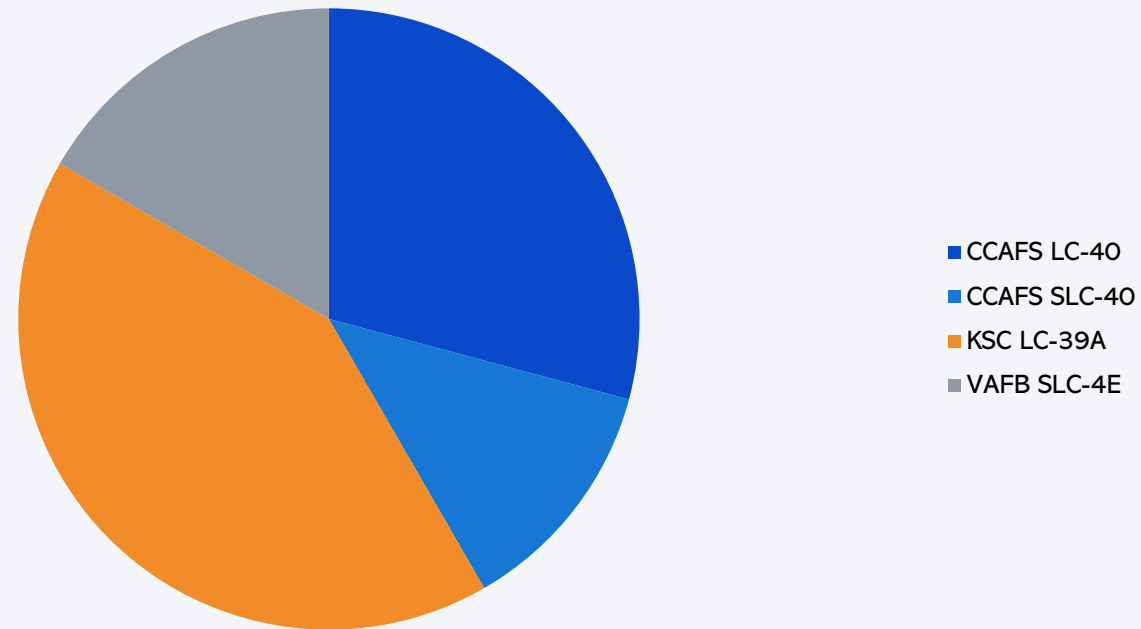


Section 4

Build a Dashboard with Plotly Dash

Dashboard sample: successful landings by site

The 56-row dashboard sample has 24 successful landings.



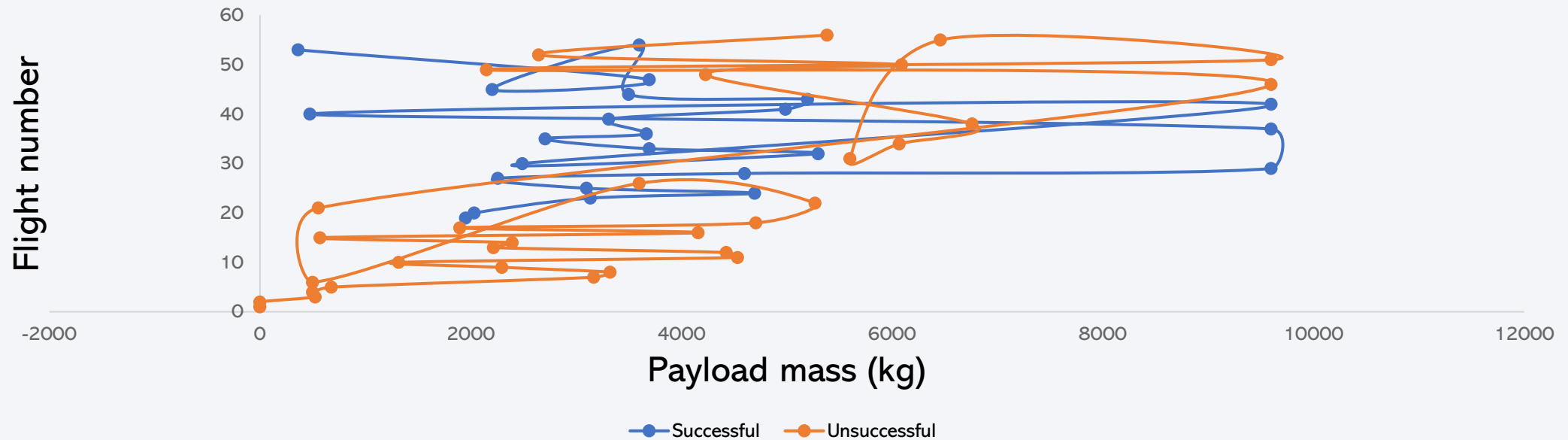
Dashboard sample: KSC success ratio

KSC LC-39A has the highest sample rate: 10/13 (76.9%).



Payload mass vs. launch outcome

The dashboard sample links payload mass and landing outcome.



Section 5

Predictive Analysis (Classification)

Four-model comparison

Five-fold validation on 72 training rows; 18 launches reserved for final test.

Model	Best setting	CV	Test	Test correct
Decision tree	depth 2	87.6%	77.8%	14/18
Logistic regression	C = 0.1	85.0%	77.8%	14/18
Support vector machine	RBF, C = 1	84.9%	77.8%	14/18
K-nearest neighbors	k = 3	76.5%	72.2%	13/18

Decision tree is the reported best model: it shares the top test score and has the highest cross-validation score (87.6%).

Decision-tree test errors

Confusion matrix for 18 held-out launches (rows = actual, columns = predicted).

	Predicted 0	Predicted 1
Actual 0	5	1
Actual 1	3	9

Accuracy: $14/18 = 77.8\%$.

One failed landing was predicted as successful; three successes were missed.

Logistic regression and SVM also scored 77.8%, with three false-success predictions each.

Conclusions

Landing success increased over time in this historical sample, while launch-site and orbit groups also differ in composition.

KSC LC 39A had the highest observed site rate (17/22, 77.3%); GTO landings were less frequent (14/27, 51.9%) than VLEO (12/14, 85.7%).

The decision tree reached 77.8% test accuracy and 87.6% five-fold validation. Its small test set and one false-success prediction call for more data before operational use.

Folium shows the VAFB pad about 1.35 km from the mapped coastline and 1.27 km from a main railway. These are geographic associations, not evidence that proximity causes landing success.

Sources and reproducibility

Project notebooks, code and interactive maps: <https://github.com/delt-kz/ibm-spacex-capstone-2026>

Launch, SQL and dashboard datasets: IBM Skills Network Applied Data Science Capstone; source links are in the repository notebooks.

Map baselayer and nearby coastline, railway and road geometries: OpenStreetMap via Overpass API (28 July 2026 snapshot). Distances are straight-line estimates to mapped way segments.

AI assistance was used for analysis and report drafting. Review the work before academic submission.

Thank you!

IBM Skills Network datasets · Historical sample through 2020 · AI-assisted draft

